

CHAPTER 2: ORGANIZING AND SUMMARIZING DATA

[This chapter is based on Sections 2.1-2.2 from the textbook]

2.1 Organizing Qualitative Data

In this section we will learn how to organize and summarize qualitative data. Usually, when we think of graphs and summary tables, we think also of numbers. However, as you may recall, qualitative data represents characteristics or measures to categorize an individual. Then, we need to represent these categories in our summaries and graphs. We start with the following definition.

Definition 2.1.

- (i) A frequency is the number of observations in each category
- (ii) A frequency distribution lists each category of data and the number of occurrences for each category of data.
- (iii) A relative frequency is the proportion (or percent) of observations within a category and is found dividing the frequency of each category by the sum of all frequencies, that is, divided by the sample size.
- (iv) A relative frequency distribution lists each category of data together with the relative frequency.

The frequency only says how many observations we have of each category, so it does not give any information about each of the categories with respect to the rest of the sample. The relative frequency shows the proportion of the sample that is represented by each category. However, it does not represent the size of the sample.

Let's see an example.

Example 2.1. A physical therapist wants to determine types of rehabilitation by her patients. To do so, she obtains a simple random sample of 30 of her patients and records the body part requiring rehabilitation in Table 1.

Back	Back	Hand	Wrist	Back	Groin
Back	Shoulder	Shoulder	Hip	Knee	Hip
Neck	Knee	Knee	Shoulder	Shoulder	Back
Back	Back	Back	Knee	Knee	Back
Hand	Back	Wrist	Elbow	Back	Back

- (a) Construct a frequency distribution of location of injury
- (b) Construct a relative frequency distribution of location of injury

Solution.

- (a) To construct a frequency distribution, we count how many repetitions of each category we have in the sample. In this case, we obtain the following table:

Category	Frequency
Back	12
Elbow	1
Groin	1
Hand	2
Hip	2
Knee	5
Neck	1
Shoulder	4
Wrist	2

To double check that our count is correct, we should make sure that the sum of all the frequencies equals the

sample size. In this case, we sum all the numbers on the right-hand side column and we notice that it adds up to 30.

- (b) To construct the relative frequency distribution we use the frequency distribution and we divide each frequency by the sample size (30). We obtain the following table:

Category	Relative Frequency
Back	$12/30 = 0.4$
Elbow	$1/30 = 0.033$
Groin	$1/30 = 0.033$
Hand	$2/30 = 0.066$
Hip	$2/30 = 0.066$
Knee	$5/30 = 0.1667$
Neck	$1/30 = 0.033$
Shoulder	$4/30 = 0.133$
Wrist	$2/30 = 0.066$

In this case, the right-hand side column should add up to 1.

□

A table is great for summarizing data, but they are frequently difficult to read and parse. An option with the same amount of information, but friendlier to our eyes are bar graphs.

Definition 2.2. A bar graph is constructed by labeling each category of data on either the horizontal or vertical axis and the frequency or relative frequency of the category on the other axis. Rectangles of equal width are drawn for each category, and the height of each rectangle represents the category's frequency or relative frequency.

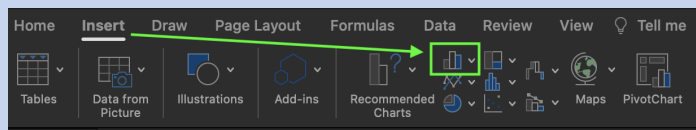
Let's continue with our example.

Example 2.2. Use the data from Example 2.1 to construct a frequency bar graph and a relative frequency bar graph.

In this case we only have a few categories, so we can draw the histograms. However, we won't be able to do this all the time. Instead, we use Excel and we follow the algorithm below. We construct the bar charts using Excel because the TI calculators do not have the ability to do them.

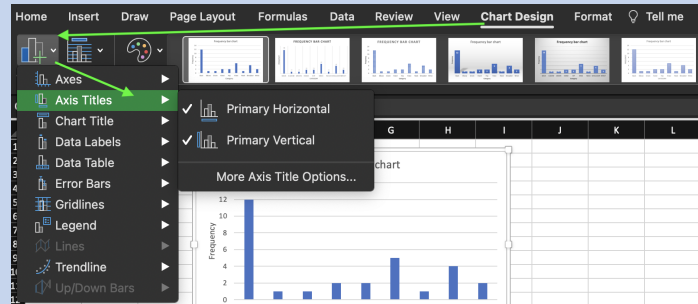
Algorithm 2.1 (Bar graphs using Excel). To construct a bar graph in Excel follow these steps:

1. Write the frequency (or relative frequency) table using one column for the categories and another column for the frequencies (or relative frequencies)
2. Select the table you constructed in Step 1
3. Click the **Insert** tab from Excel
4. Select the column graph (top left option) as shown in the figure below

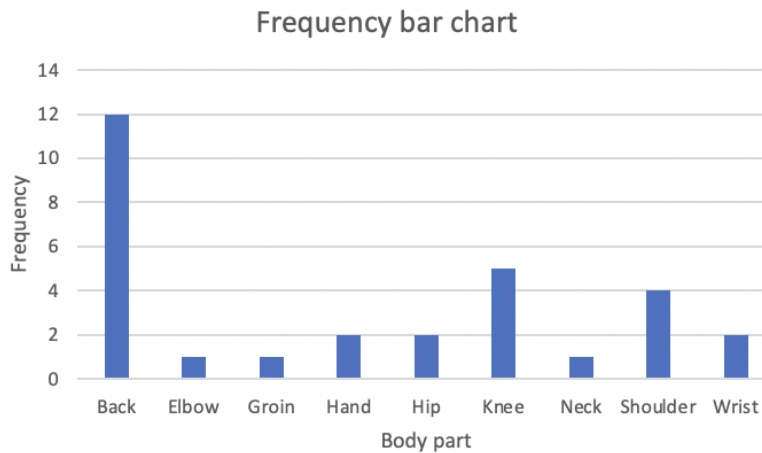


5. Select your favorite option, and the bar graph will show up in your Excel sheet
6. To add titles to the axes, select the graph you just inserted and follow these steps:
 - 6.1 Go to the **Chart design** tab
 - 6.2 Click on **Add elements** on the top left corner

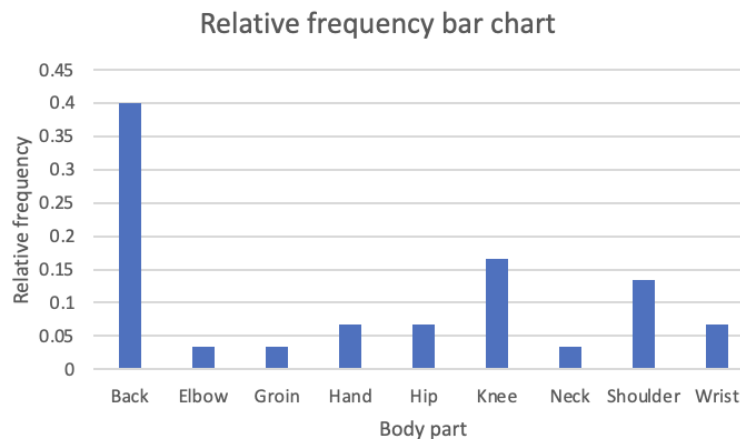
6.3 Select Axis titles and the horizontal or vertical title option, as shown in the figure below



Solution. We follow the steps from the algorithm above, and obtain the following frequency chart:



Similarly, we obtain the following relative frequency chart:



Observe that both plots are exactly the same, except by the numbers on the vertical axis. This is expected, since we are graphing the same data under two different representations.

The Excel sheets I used to construct these graphs are available on Blackboard. □

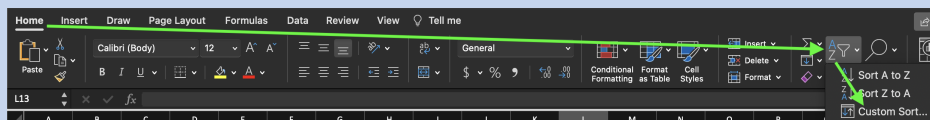
Since we cannot naturally order the categories in qualitative data, in general the order of the bars does not matter. However, organizing them in decreasing order can be helpful to make decisions more easily.

Definition 2.3. A Pareto chart is a bar graph whose bars are drawn in decreasing order of frequency or relative frequency.

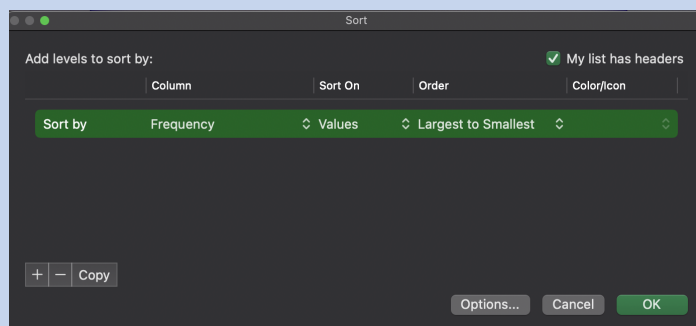
To build a Pareto chart we follow the steps from Algorithm 2.1 and sort the frequency (or relative frequency) in decreasing frequency values, as indicated in the algorithm below.

Algorithm 2.2 (Pareto chart using Excel).

1. Follow Step 1 from Algorithm 2.1
2. Select the table
3. In the Home tab of Excel, click on Sort & filter, and select Custom sort as shown in the following picture:

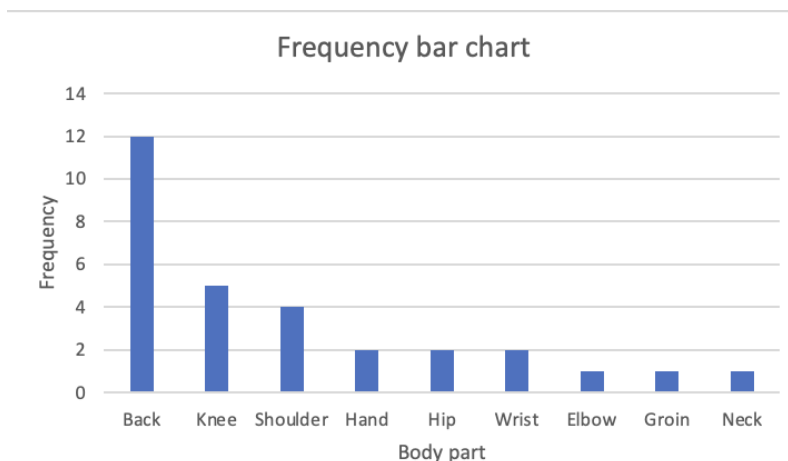


4. Select organize in decreasing value by frequency, as shown in the following picture:



5. Follow steps 2-6 from Algorithm 2.1

The graph below is a Pareto chart of the frequency table from Example 2.1



Bar graphs can also be used to compare two samples, as in the example below.

Example 2.3 (Exercise 17 from textbook). A survey of U.S. adults and teens (ages 12-17) was administered by Pew Research, to determine the number of texts sent in a single day. The results from the survey are presented in the following table:

<i>Number of texts</i>	<i>Adults</i>	<i>Teens</i>
<i>None</i>	<i>173</i>	<i>13</i>
<i>1-10</i>	<i>978</i>	<i>138</i>
<i>11-20</i>	<i>249</i>	<i>69</i>
<i>21-50</i>	<i>249</i>	<i>113</i>
<i>51-100</i>	<i>134</i>	<i>113</i>
<i>101 or more</i>	<i>153</i>	<i>181</i>

(a) Construct a relative frequency distribution for adults

(b) Construct a relative frequency distribution for teens

(c) Construct a side-by-side relative frequency bar graph

(d) Comment on the texting habits of adults and teens

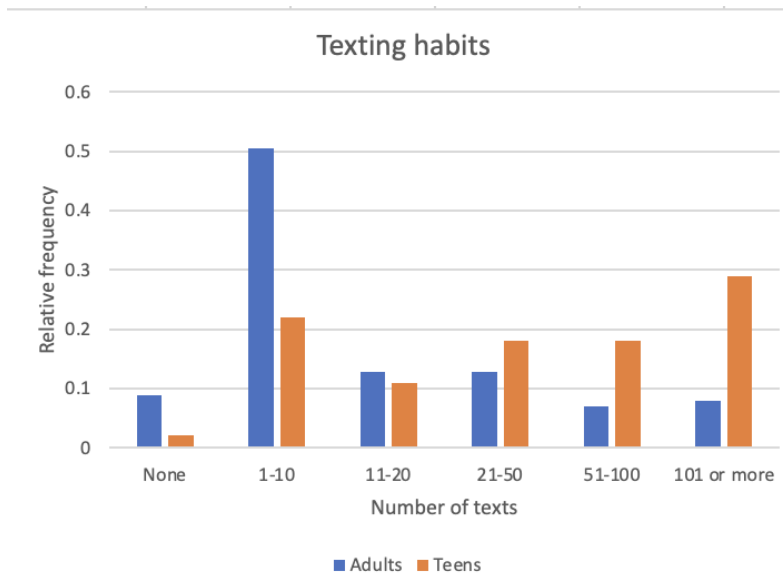
Solution. To solve parts (a) and (b) we compare adults against adults and teens against teens. That is, to compute the relative frequency of each category of adults we divide by the total number of adults that participated in the study, and similarly with teens.

Summing over the adults column, we obtain that 1936 adults participated. Similarly, we obtain that teens 627 participated. Observe that since the number of adults that participated is over 3 times the number of teens, studying the relative frequencies makes more sense than the frequencies.

Hence, the relative frequency tables are:

Number of texts	Adults	Teens
None	$173/1936 = 0.089$	$13/627 = 0.021$
1-10	$978/1936 = 0.505$	$138/627 = 0.220$
11-20	$249/1936 = 0.129$	$69/627 = 0.110$
21-50	$249/1936 = 0.129$	$113/627 = 0.180$
51-100	$134/1936 = 0.069$	$113/627 = 0.180$
101 or more	$153/1936 = 0.079$	$181/627 = 0.289$

Following the steps from Algorithm 2.1 with two columns of data, we obtain the following chart:



Observe that we now plot two samples of data, so we need to add a legend telling us what are the colors representing. In this case, blue represents adults and orange represents teens.

From the chart, we observe that a large percentage of adults texts a little bit every day. As the number of

texts/day increases, the number of adults decreases. If we focus on ≤ 20 texts/day, adults are get the largest frequency and the opposite happens with ≥ 21 . That is, teens text a lot more than adults. $\square$

Observe that, even though the number of texts a person sends in one day is quantitative, in the survey the participants were asked for ranges of texts. Hence, the data becomes qualitative.

The last representation of qualitative data that we will learn is a pie chart.

Definition 2.4. A pie chart is a circle divided into sectors. Each sector represents a category of data and the area of each sector is proportional to the frequency of each category.

Let's see an example.

Example 2.4. The data presented in the table below represent the educational attainment of residents of the United States 25 years or older in 2013, based on data obtained from the U.S. Census Bureau. The data are in thousands.

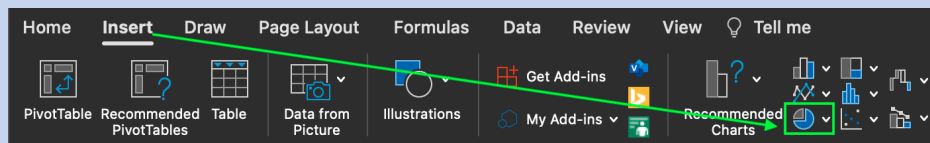
Educational attainment	Relative Frequency
Not a high-school graduate	24,517
High school diploma	61,704
Some college, no degree	34,805
Associate's degree	20,367
Bachelor's degree	41,575
Graduate or professional degree	23,931
Total	206,899

Construct a pie chart of the data.

To construct a pie chart with Excel, we use the following algorithm.

Algorithm 2.3 (Pie chart with Excel).

1. Follow steps 1, 2 and 3 from Algorithm 2.1
2. Select the pie chart (bottom left option) as shown in the figure below



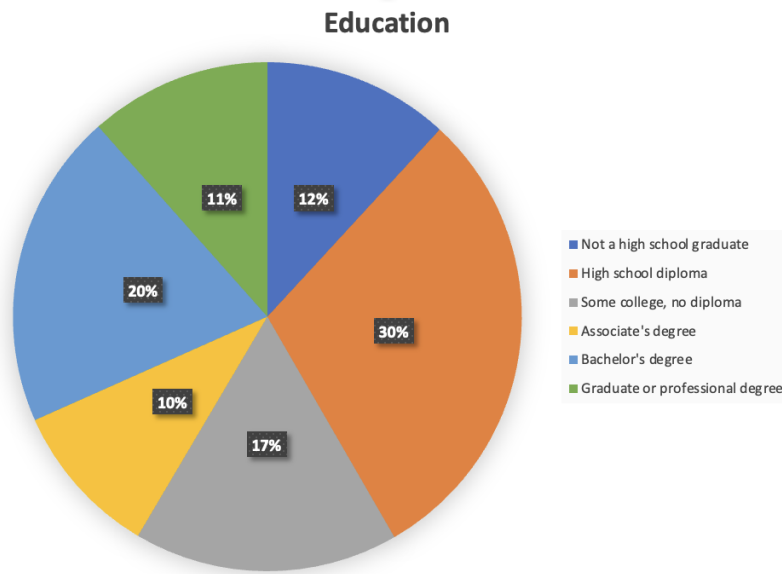
3. You may use the Chart Design tab and choose the chart style that you prefer.

We now use Algorithm 2.3 to solve the example.

Solution. We first construct the relative frequency table, dividing each of the frequencies by the sample size 206,899. We obtain:

Educational attainment	Frequency
Not a high-school graduate	0.118
High school diploma	0.298
Some college, no degree	0.168
Associate's degree	0.098
Bachelor's degree	0.201
Graduate or professional degree	0.116
Total	1

We construct the pie chart following the steps from Algorithm 2.3 using relative frequencies and obtain:



□

Pie charts are useful to represent the proportion that each category of a sample uses. However, it is not recommended when comparing two samples because comparing the areas is not intuitive.

2.2 Organizing Quantitative Data

Organizing qualitative and quantitative data is different, but there are plenty of similarities. When we organize qualitative data, each observation is a category and we build frequency distributions and charts that represent how often we observe each category.

When we organize quantitative data, instead, the categories are not always obvious and, hence, the first step is to build them. These categories are called classes. If there are only a few different values, we may use each observation as a category. However, if there are many possible values, we may group them and summarize the data using these groups as categories.

Histograms with discrete data

Once we have the categories (or classes) we can build the frequency and relative frequency distributions. Let's consider an example with only a few values of the variables.

Example 2.5 (Example 1 of the textbook). *The manager of a Wendy's fast-food restaurant wants to know the typical number of customers who arrive during the lunch hour. The data in the table below represent the number of customers who arrive at Wendy's for 40 randomly selected 15-minute intervals of time during lunch.*

7	6	6	6	4	6	2	6
5	6	6	11	4	5	7	6
2	7	1	2	4	8	2	6
6	5	5	3	7	5	4	6
2	2	9	7	5	9	8	5

Construct a frequency and relative frequency distribution of the data.

Solution. In this case the values go from 1 to 11, so we can use each possible outcome as a category. We obtain the following frequency and relative frequency distributions:

Number of customers observed	Frequency	Relative frequency
1	1	$1/40 = 0.025$
2	6	$6/40 = 0.15$
3	1	$1/40 = 0.025$
4	4	$4/40 = 0.1$
5	7	$7/40 = 0.175$
6	11	$11/40 = 0.275$
7	5	$5/40 = 0.125$
8	2	$2/40 = 0.05$
9	2	$2/40 = 0.05$
10	0	0
11	1	$1/40 = 0.025$

□

Observe that, since we had only a few possible values for the observations, building the frequency and relative frequency distributions in the example above was exactly the same as in qualitative variables.

Similarly to the qualitative data, we can build a chart to summarize the data. We define a specific chart below.

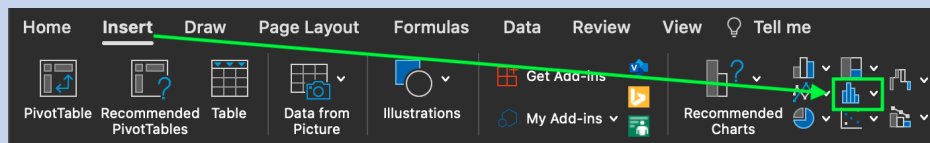
Definition 2.5. A *histogram* is constructed by drawing rectangles for each class of data. The height of each rectangle is the frequency or relative frequency of the class. The width of each rectangle is the same and the rectangles touch each other.

In simple words, a histogram is a vertical bar chart for quantitative data, where the bars must touch. The bar charts we learned in the previous section can have vertical or horizontal bars, and the bars do not touch.

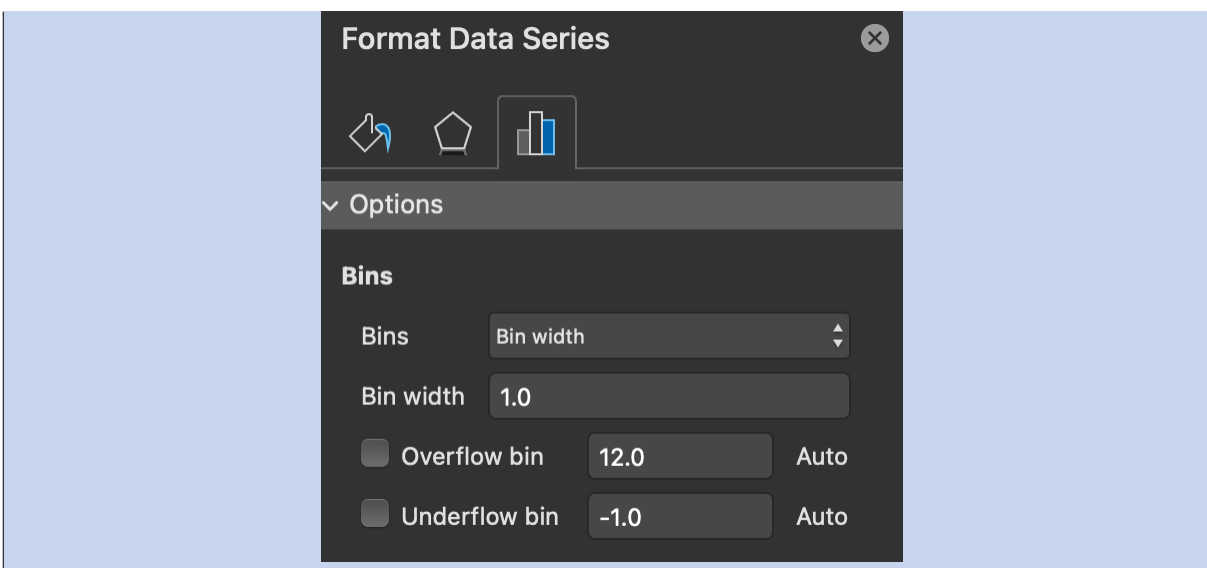
To construct histograms with Excel we have two options. In the first option, we enter the raw data as described in the following algorithm.

Algorithm 2.4 (Histogram with Excel starting from raw data).

1. Enter the raw data to Excel
2. Select all the data
3. Go to the **Insert** tab and select the histogram (option in the second row, second column) as shown in the figure below



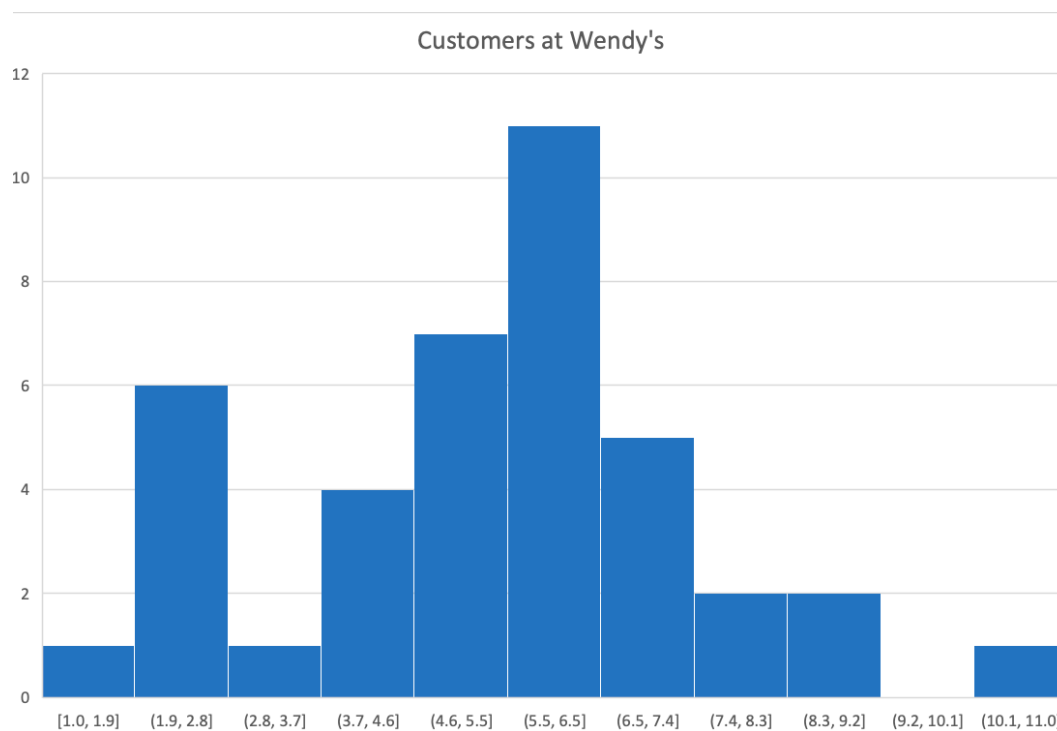
4. Click on one of the bins
5. Select the bin width on the right-hand side menu, as shown in the figure below



Let's do an example.

Example 2.6 (Example 2 from the textbook). *Construct a frequency histogram using the data from Example 2.5.*

Solution. Following the steps from Algorithm 2.4 we obtain the following frequency histogram:



□

A similar figure can be obtained with a TI83/84 calculator. The steps are provided in the following algorithm:

Algorithm 2.5 (Histogram with TI83/84 starting from raw data).

1. Enter the raw data:
 - i. Press **stat** button

- ii. Press 1 to enter the 1:Edit... menu
 - iii. Enter first observation under L1 and press enter
 - iv. Repeat step iii. for all the observations
2. Press the 2nd button and next y= to access the StatPlot menu. Observe that above the y= button says StatPlot in blue, which is the color of the 2nd button.
 3. Press 1 to enter 1:Plot1
 4. Use the arrows to place the cursor on ON and press enter
 5. Use the arrows to navigate until the histogram diagram (third after Type:) and press enter
 6. Press the 2nd button and mode to quit Plot 1 menu. Observe that above the mode button says quit in blue
 7. Press window and set the horizontal and vertical axes limits.
 - Xmin stands for the minimum value on the horizontal axis
 - Xmax stands for the maximum value on the horizontal axis
 - Xscl stands for the class width
 - Ymin stands for the minimum value on the vertical axis. Most of the time we want to set this value to 0
 - Ymax stands for the maximum value on the vertical axis.
 8. Press graph to display your histogram

Observe that the horizontal axis of our histogram above has intervals of data. This is because, in general, we group categories when we work with quantitative variables, and the best way to build these groups is as intervals. We will see how to do this in the following subsection.

Histograms with continuous data

We introduce some key concepts regarding categorization of quantitative data in the following definition.

Definition 2.6. A class or category is represented by an interval of values. The smallest value within the class is called lower class limit, the largest value within the class is called upper class limit, and the difference between consecutive lower class limits is called class width.

Observe that the class width is independent of which pair of consecutive lower class limits we consider. In other words, all the classes must be equally wide.

Let's see an example.

Example 2.7. The data in the table below represents the number of U.S. citizens with ages 25-74 who had earned a bachelor's degree or higher in 2013.

Age	Number (in thousands)
25-34	14,481
35-44	14,155
45-54	13,802
55-64	12,121
65-74	7,009

For each category, indicate the lower and upper class limit. Additionally, compute the class width.

Solution. We obtain:

Class	Lower class limit	Upper class limit
25-34	25	34

35-44	35	44
45-54	45	54
55-64	55	64
65-74	65	74

and the class width is

$$35 - 25 = 10$$

In the computation of the class width we chose to subtract the lower class limit of the first and second categories. However, we can choose any pair of consecutive lower limits. For example, if we choose the fourth and fifth category, we would obtain $65 - 55 = 10$. $\square$

When we select our classes, it is key to choose classes that do not overlap. Otherwise, we will get very confused when deciding where a specific observation belongs, and there will be some observations counted more than once.

In the example above, we were studying people with age between 25 and 74. However, we may not define the lower and upper limit of the population. In such case, we have an open ended table.

Definition 2.7. A table is *open ended* if the first class does not have a lower class limit or if the last class does not have an upper class limit. If a table is open ended, the class width cannot be computed using the lower class limit of the first class.

Example 2.8. The data in the table below represents births to unmarried mothers in 2012 in the US.

Age	Number
10-19	246,561
20-29	994,828
30-39	336,179
40 and over	28,075

Observe that the table is open ended above, as there is no upper class limit for the last class. The class width is 10, and we can compute it using any pair of consecutive lower class limits.

When we choose the classes, we need to keep in mind that we want to summarize the information efficiently. That is, we don't want to use too many categories because we will end up having (approx) one category per observation. On the other hand, we don't want to use only a few categories because grouping the observations in very large intervals will hide some of the key observations we need for our study.

There is no absolute rule to select categories, and each case is different. However, the following guidelines might be helpful:

- **Choosing the lower class limit of the first class:**

Choose the smallest observation or a convenient number that is slightly lower than the smallest observation

- **Determining the class width:**

We first decide the number of classes (usually 5-20). The smaller the data set, the fewer classes we should have. Once we have the number of categories, we compute the class width as follows:

$$\text{Class width} \approx \frac{\text{Largest data value} - \text{Smallest data value}}{\text{Number of classes}}$$

The approximation sign ($\approx$) means that the computation is a guideline, and we can round the value up to a convenient value.

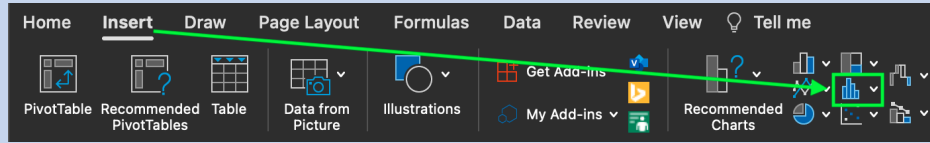
The second option to build histograms with Excel is using the frequency or relative frequency distribution as data. In this case, we use the following algorithm.

Algorithm 2.6 (Histogram with Excel starting from frequency or relative frequency distribution).

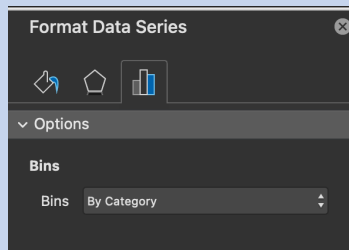
1. Write the frequency (or relative frequency) distribution using one column for the categories and one for the frequency (or relative frequency)

2. Select the table you constructed in Step 1

3. Go to the **Insert** tab and select the histogram (option in the second row, second column) as shown in the figure below



4. Double-click on one of the bars from the histogram and select “by category” from the drop-down menu as shown in the following figure



Let's see an example.

Example 2.9. The table below represents the five-year rate of return in percent for a simple random sample of 40 large-blended mutual funds. Construct a frequency and relative frequency distribution of the data.

10.94	14.60	12.80	16.00	11.93	15.68	9.03	13.40
10.53	13.98	13.86	12.36	13.54	9.94	13.93	13.63
14.12	14.88	14.77	13.13	8.28	19.43	12.98	13.16
12.26	14.20	14.80	13.26	13.67	10.08	14.86	8.71
12.17	10.26	15.22	13.36	13.55	13.90	15.64	12.80

Solution. To construct the frequency and relative frequency distribution, let's first decide on the lower class limit of the first class and the class width.

Observe that 8.28 is the lowest value of the sample. To make things easier, let's set 8 as the lower class limit of the first class.

To compute the class width, let's try to get 12 classes. Then, the class width is:

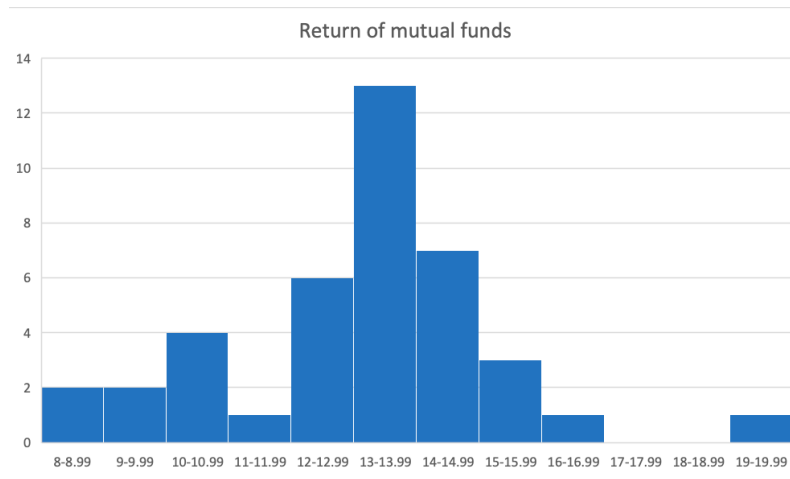
$$\begin{aligned}\text{Class width} &\approx \frac{\text{Largest data value} - \text{Smallest data value}}{\text{Number of classes}} \\ &= \frac{19.43 - 8.28}{12} = 0.929\end{aligned}$$

Since 0.929 is a difficult number to work with, let's round it up and work with a class width of 1. Then, we obtain the following frequency and relative frequency distribution:

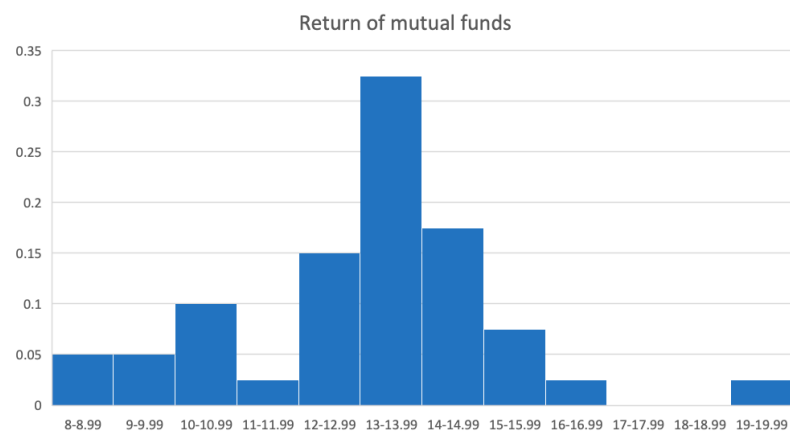
Class	Frequency	Relative frequency
8-8.99	2	$2/40 = 0.05$
9-9.99	2	$2/40 = 0.05$
10-10.99	4	$4/40 = 0.1$
11-11.99	1	$1/40 = 0.05$
12-12.99	6	$6/40 = 0.15$
13-13.99	13	$13/40 = 0.325$
14-14.99	7	$7/40 = 0.175$
15-15.99	3	$3/40 = 0.075$
16-16.99	1	$1/40 = 0.05$
17-17.99	0	0

18-18.99	0	0
19-19.99	1	$1/40 = 0.05$

Following the steps from Algorithm 2.6, we obtain the following histogram of the frequency



and the following histogram for the relative frequencies



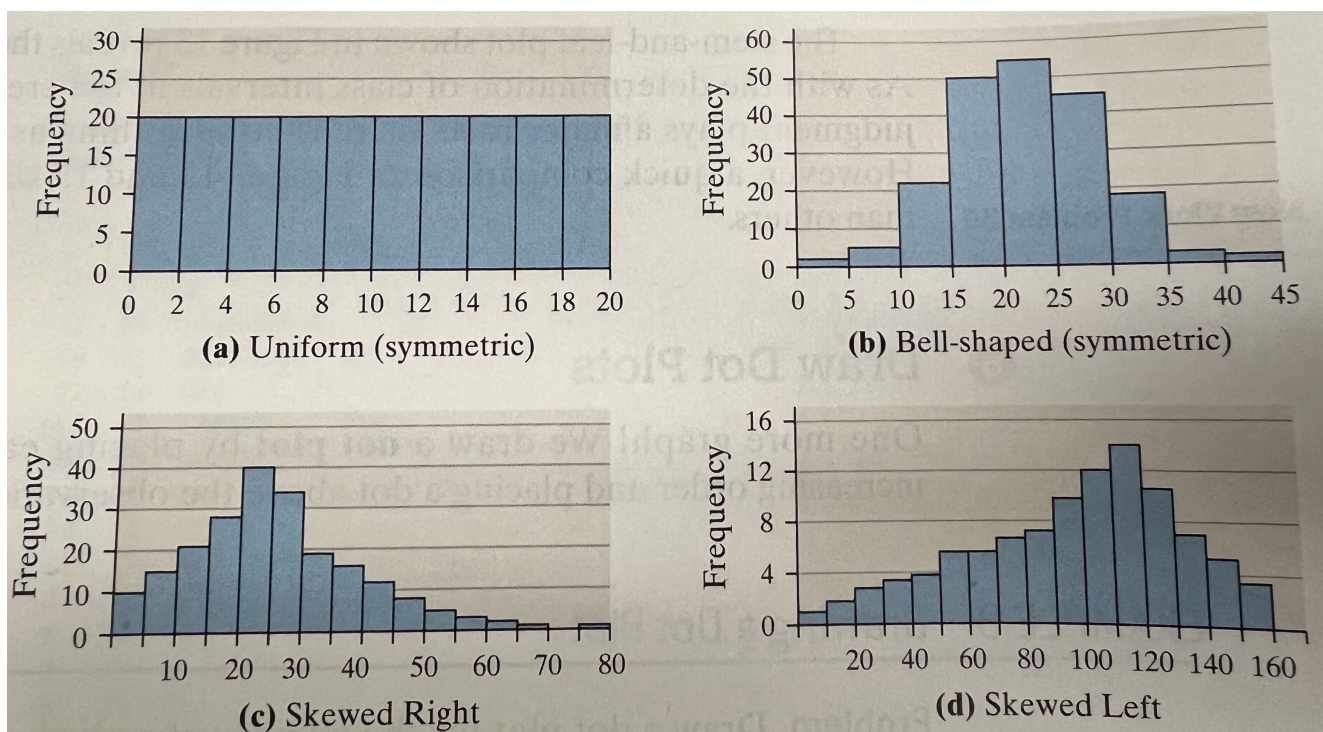
Observe that the histograms look very similar in shape. The only difference is on the vertical axis numbers. In both cases, observe that the horizontal axis has the categories listed.

□

Understanding the shape of a histogram

The shape of the histogram tells us a lot about the data. There are three types of shapes: symmetric, skewed to the right, and skewed to the left. The following figure¹ shows all these shapes:

¹Figure 15 from chapter 2 in the textbook



A histogram is:

- Symmetric if a vertical line at the center of the histogram works as a mirror, that is, we see approximately the same on the right and left-hand side. In the figure, panels (a) and (b) are symmetric
- Skewed right if the right side of the peak is longer than the tail to the left side. Panel (c) in the figure is skewed right.
- Skewed left if the left side of the peak is longer than the tail to the right side. Panel (d) in the figure is skewed left.

Stem-and-Leaf plots

Stem-and-leaf plots are an alternative to represent quantitative data, where we split the digits of each observation to group them according to their common digits from left to right. The leaf of an observation is the last digit on the right-hand side, and the stem is the rest of the number. The main steps are described below:

1. Write a column with all the stems and a vertical line on the right-hand side
2. On the right-hand side of each stem, write the remaining digit corresponding to each observation. Write them one next to the other, in ascending order.

When we choose the stems, we may choose 1, 10, 100, 0.1, 0.01, etc. Let's see an example.

Example 2.10 (based on exercise 32 from textbook). *The volume of a stock is the number of shares traded on a given day. The following data (in millions) represents the volume of Altria Group stock traded for a random sample of 35 trading days in 2014.*

6.4	23.6	18.9	7.8	7.8
8.5	9.0	14.8	14.4	8.6
6.4	10.3	10.1	10.9	11.2
13.6	9.1	7.8	15.3	14.0
7.8	7.9	17.1	16.6	7.7
7.7	10.2	10.5	8.4	7.9
10.9	20.1	8.9	15.4	8.3

Build a stem-and-leaf plot of the data

Solution. We build a stem-and-leaf plot where the stems are the left-hand side digits up to the unit. Then, the leafs represent the decimal points. For example, 6.4 is represented as 6|4, and if we want to represent the observations 10.1 and 10.9 we write 10|19.

Since the data goes from 6.4 to 23.6, we obtain the following plot:

6		44
7		7788899
8		3456899
9		01
10		123599
11		2
12		
13		6
14		048
15		34
16		6
17		1
18		9
19		
20		1
21		
22		
23		6

□

Stem-and-leaf plots are similar to histograms because we need to pick categories, and we will easily see which categories are more popular and how much. The difference is that stem-and-leaf plots contain all the information, that is, from a stem-and-leaf plot I can recover all the data. From a histogram, instead, we can only retrieve a summary of the data. Another difference is that in stem-and-leaf plots, the class width must be 1, 10, 100, etc because we are using decimals to summarize the data. In histograms, instead, we have more freedom to pick the class width.

Choosing histogram or stem-and-leaf plot

Now that we've learned how to construct histograms and stem-and-leaf plots, how do you decide which one you want to use?

As we discussed above, both are great at giving us a visual idea of the data. However, stem-and-leaf plots display each observation explicitly, whereas histograms show us the frequency of each category only. Also, in histograms we have more freedom to choose the classes.

The answer to 'which one is better' depends on each case and what we want to do with the data. But as a rule of thumb, we usually prefer stem-and-leaf plots if the sample is small. Otherwise, the amount of information becomes overwhelming. For large samples, histograms tend to be better.